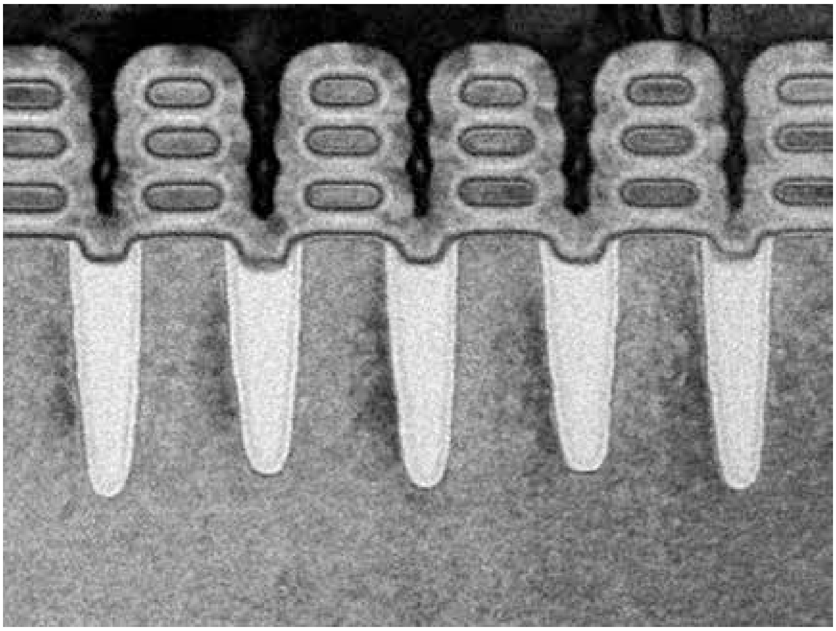
Credit: IBM

Nanosheet Transistor

would be required to push the limits further. Mainstream companies are expected to get into risk production of 2nm around 2024 to 2029. And whatever comes after that will take even longer to perfect and make viable. There's an additional problem of scaling below 2nm, and it's an obvious one, that of cost. Current gen process nodes such as 7nm have cost upwards of 9-10 billion dollars to perfect. And the ones that follow will cost even more. While using Silicon is seeming to be more and more difficult, manufacturers still prefer Silicon because the decades of research and development into silicon has made it a mature material to work with. Switching to another material would require monumentally greater investments to bring it up to the same level as Silicon.

So what are researchers working on to replace silicon? A variety of materials such as black phosphorus, graphene, boron nitride nanosheets and several others are being explored. Many of these are termed 2-D materials because they can be made into layers that are only an atom or two thick. Materials science pertaining to such materials have since long theorised their ability to be drawn into such thin sheets but the techniques required to make it possible were only invented in the early 2000s.

These new technologies have already been implemented in devices around us such as night-vision sensors used in certain cameras. We haven't yet managed to devise methods that would allow high-volume production using such materials. Worry not, chip manufacturers such as IBM, TSMC, GlobalFoundries and Intel are researching different methods to make this possible.

Graphene is yet another material that has shown immense promise. It can be manufactured in layers that are just one atom thick, however, it is unlikely to replace silicon because its chemical and physical properties are not similar to those of silicon. However, when combined with other materials, it does show some promise. Bio-gated transistors are one such application. It uses biologically activated molecules to graphene and that is, in turn, attached to silicon. If you're a fan of *Star Trek* then you're probably thinking of bio-neural gel packs, a form of circuitry that was much faster than traditional circuits. While we'd all like technology such as bio-neural circuitry to become a reality, the applications being worked in the labs are in no way similar.

Another material being explored is Gallium Nitride. It too has already been used in commercial applications. Gallium Nitride has a much wider band gap so it can handle much higher voltages without causing it to jump across gaps. As a result, transistors can be packed together in a much denser manner. Currently, the transistors aren't being used to make processors but it won't be long before it's considered to be a viable candidate. As of now, it's used in power electronics i.e. electric chargers for devices such as laptops and mobiles. You'll see these being marketed as GaN devices.

PACKAGING

And what happens if you've already pushed the boundaries of material science? Well you can increase the size of the processors. Processor chips can grow longer in length and width but that's going to cause heating problems and manufacturers will have to incorporate even bigger heatsinks and cooling systems. The other way is to increase performance by leveraging different packaging technologies. Manufacturers such as Intel and TSMC have already demonstrated packaging methods that allow chips to be stacked on top of each other. TSMC's implementation is called Chip-on-Wafer-on-Substrate or CoWoS. They aren't stacking processor cores upon each other because that would compound the heating issues. As of now they're playing around with stacking cache memory onto the memory substructure to allow CPU cores to gain access to larger amounts of cache memory. Chips such as NVIDIA's Pascal and Volta, Intel's Spring Crest and AMD's Vega GPUs all use CoWoS.

CoWoS was succeeded by another stacking technology which allowed direct bonding i.e. silicon-onto-silicon without any intermediate material or substrate. It is called Direct Bond Interconnect or DBI. It is used in camera sensors such as Sony's IMX260 but recently we've seen it being used in AMD's prototype processor that they showcased at this year's COMPUTEX in the form of AMD 3D V-Cache.

So once processors get smaller in size due to shrinking process nodes, there will be a point where increasing